Akanksha Sachan

Contact

PhD Candidate in Computational Biology
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Machine learning for dynamic gene regulation and cellular reprogramming

Education

Joint Carnegie Mellon–University of Pittsburgh PhD Program *'22 – Present*
PhD Candidate in Computational Biology *Expected '27*
University of Pittsburgh School of Medicine ('24 – Present)
Advised by Dr. Jishnu Das
Carnegie Mellon University ('22 – '24)
Kenneth C. Griffin School of Computer Science | GPA: 3.58/4

Indian Institute of Technology (IIT), Bombay *'18 – '22*
BTech. in Chemical Engineering | GPA: 8.06/10

Awards and Honors

American Federation for Aging Research (AFAR) Graduate Scholarship *'25*

Best Poster Award at the Aging Institute Research Day, UPitt *'24*

1st Place, Student A100 track NeurIPS LLM Efficiency Challenge | team of 5 *'23*

$500 Track Prize (Healthcare Accessibility for Underrepresented Communities), Pitt Challenge Hackathon (6th annual) | team of 3 *'22*

Recipient of **Undergraduate Research Award** (URA-01) by IIT Bombay *'22*

Recipient of Merit-cum-Means (MCM) Scholarship by IIT Bombay for two semesters for financial hardship *'21*

Top 0.2% (Rank 3188 out of **1.1M+** candidates) in JEE (Main) *'18*

Top 1.5% (Rank 2517 out of **160k+** candidates) in JEE (Advanced) *'18*

Invited Talks

"Interpretable machine learning coupled to gene regulatory networks uncovers subcircuits underlying cell fate decisions" | *preprint*

International Society for Computational Biology (ISCB)/(ISMB) *Jul '26, '25*
Keystone Symposia: Artificial Intelligence in Molecular Biology *Sep '25*
American Association of Immunologists (AAI) Annual Meeting 2026 *Apr '26*
Center of Lung Aging and Regeneration, School of Medicine, UPitt *Nov '25*
30th Annual School of Medicine Graduate Student Research Symposium, UPitt *Oct '25*
Joint CMU-UPitt Ph.D. Program in Computational Biology, Orientation Retreat *Aug '25*

"Sex-dimorphic DNA lesion accumulation drives a PDK4 metabolic checkpoint in skeletal muscle" | *in-review at Nature Aging*

Center for Systems Immunology Annual Retreat, UPitt *Nov '25*
23rd Annual Department of Immunology Scientific Retreat, UPitt *Sep '25*
Aging Institute Research Day, UPitt *Nov '24*

"Micro-MPA digital twin, with sclerostin-dependent RANKL production reproduces pivotal dual-action effect of romosozumab found in clinical trials"

50th European Calcified Tissue Society (ECTS) Congress *Apr '23*

Research Experience Prior to Thesis

Mapping the epigenetic landscape of single-cells using 3-D chromatin architecture *'22 – '24*

Supervisor – Prof. Jian Ma, Computational Biology Department, SCS, CMU

Developed a framework to identify cis-regulatory 3-D genomic hubs controlling target-gene expression and key transcription factors in healthy and cancerous mouse-brain single-cell, multi-omic profiling.

Translational Mechanobiology Lab, MBI, National University of Singapore *'22*

Summer Internship | Supervisor – Prof. Hanry Yu, Dept. of Physiology, YLL School of Medicine

Built a Generative-Adversarial-Network (GAN) pipeline for colorectal-cancer imaging to support clinical deep-learning deployment

Lab for Bone Biomechanics, Institute of Biomechanics, ETH Zurich *'21*

Summer Internship | Supervisor – Prof. Ralph Muller, Dept. of Health Sci&Tech

Added a Type-2 Diabetes pathway to an agent-based osteoporosis model and ran large-scale simulations on the CSCS Swiss National Supercomputing Center

Multicellular Systems Engineering Lab, University of Notre Dame *'20*

Summer Internship | Supervisor – Prof. Jeremiah Zartman, Chemical & Biomolecular Engineering

Built a morphometric feature-extraction pipeline for *Drosophila* wing imaginal discs (OpenCV, Fiji) to study organism-wide cell-signaling

Key Coursework

Machine Learning & Statistics: Intermediary Statistics (CMU: 36-705), Probabilistic Graphical Models (CMU: 10-708), Introduction to Machine Learning (CMU: 10-701)

Predicting genomic-hub mediated TF-perturbation response via geometric deep learning *'23*

10-708: Probabilistic Graphical Models

Augmented a GNN for Perturb-seq response prediction with a Hi-C contact graph (Cancer cell-line), adding 3D-genome topology to the co-expression and Gene Ontology graph features.

Single-cell DNA methylation prediction using single-cell HiC data and Graph NNs *'23*

02-710: Computational Genomics

Developed a **GNN** to predict methylation from HiC data, connecting structure to phenotype; clustered **single-nucleus DNA methylation** data with K-means and t-SNE.

Athletic Accolades

Bronze Medal | 35th Intercollegiate (inter-IIT) Aquatics Meet | 3rd in Medley-Relay *'19*

3rd Position | Comp-Bio Department, Pretty Good Race (5km) for SCS, CMU *'23*

Swam 12-hours | Swimathon organized by the IIT Bombay aquatics committee *'19*

Ran 26.2 miles | Dick's Sporting Goods Pittsburgh Marathon | 5 hours 15 mins *'24*

Ran 13.1 miles | Dick's Sporting Goods Pittsburgh Half Marathon | 2 hours 30 mins *'23*

Leadership & Societal Contributions

President — *Jun '26 – Present*
Allegheny Science Policy and Governance (ASPG)
Previously Vice President (Jul '25 – Jun '26). Leading a recognized chapter of the **National Science Policy Network**; advocating for **AI safety and biosecurity** with Pennsylvania state legislators.

Technights Coordinator — *Aug '23 – Aug '25*
Carnegie Mellon University, Robotics Institute
Organized and led a **computer-science outreach** program teaching middle-school girls sessions on foundation models, and deep learning.

NIH Consortium Hackathon, Co-organizer & Team Lead — *Mar '24*
4D Nucleome, Seattle, Washington
Led a team of 5 to build integrative pipelines for single-cell HiC analysis from cell-atlas-scale datasets

Reviewer
ICML Foundation Model for Life Sciences Workshop 2026, *PNAS Journal* (*assisted PI*), *Aging Cell Journal* (*assisted PI*)

Teaching Assistant — *Fall '23 – Spring '24*
02-760: Lab Methods for Computational Biologists, SCS, CMU
Taught a lecture on **genomics technologies** to a class of ∼20 students, revamping the RNA sequencing module of the course.

Treasurer • Senator — *Aug '23 – Aug '25*
Graduate Student Association (*Joint CMU-UPitt Comp-Bio PhD Program*)

Editorial Board Member • Design Convenor — *'19 – '21*
Insight, the student newsletter of IIT Bombay

Convenor, Girls Swim Team — *'19 – '20*
Aquatics, Indian Institute of Technology Bombay

Consortiums & Workshops

NIH Consortium Annual Meetings — *'23, '25*
IGVF, SenNet:Cellular Senescence Network, 4D Nucleome

Grace Hopper Celebration — *Oct '24*
AnitaB.org, Philadelphia

Computational Genomics Summer Institute — *Jul '23*
University of California, Los Angeles

Grad Cohort for Women — *April '23*
Computing Research Association (*CRA-WP*), *San Francisco*

Compute Allocation Grants

ACCESS (NSF Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support) **1.5M** credits on the NCSA Delta GPU supercomputer — *'26*